

Predator-Prey Ecosystem Simulation

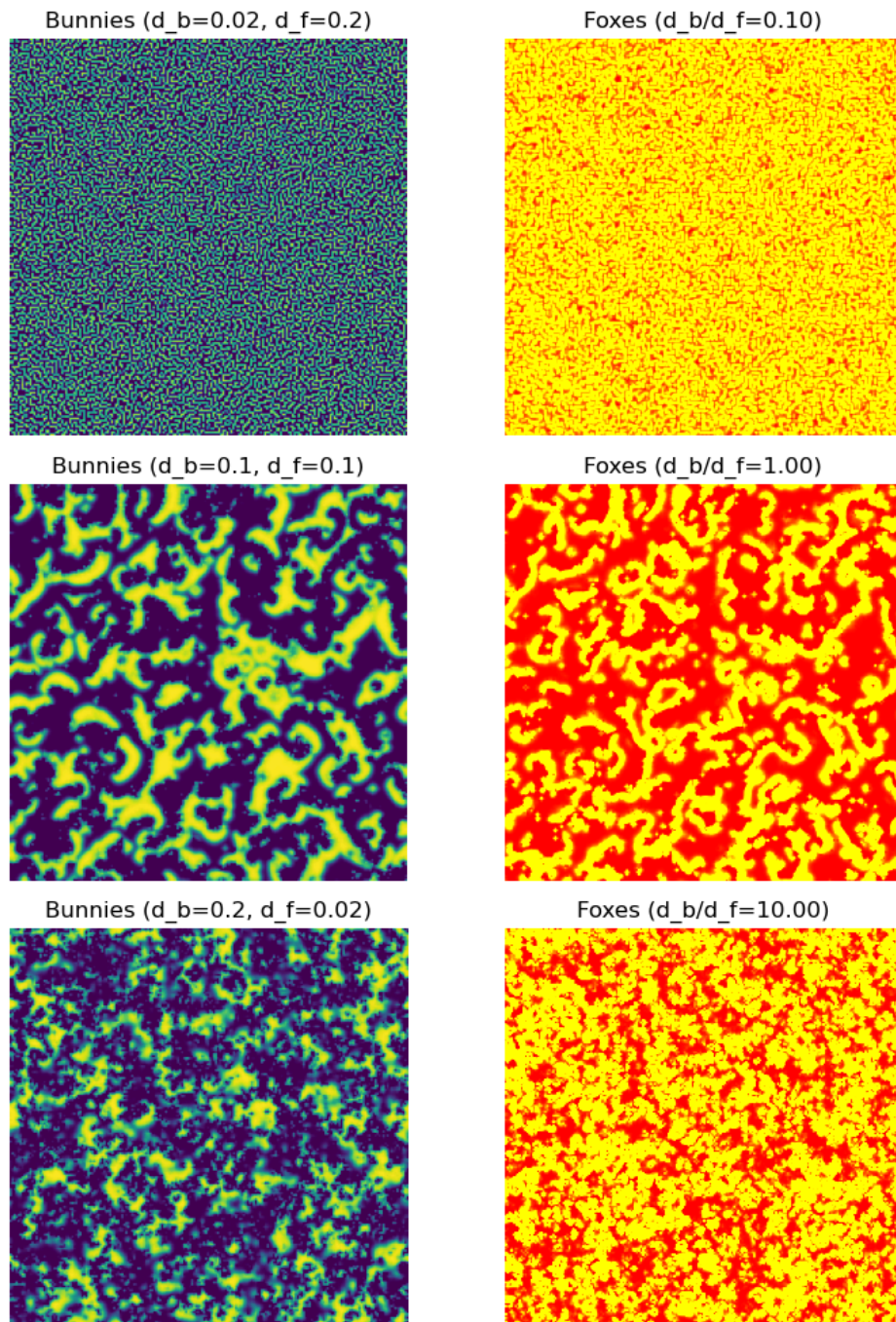
Minerva University

CS166

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Predator-Prey Ecosystem Simulation.



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Figure 1: Snapshot of simulation results.

Task 2: Empirical Analysis

Basic behavior of the model

Figure 1 shows representative snapshots of the system under different diffusion ratios. Starting from random initial conditions, the model self-organizes into spatial predator/prey patterns through the interaction of local growth, predation, and diffusion. I intentionally use the same diffusion rule for both bunnies and foxes (four-neighbor redistribution) so it follows closely with “in pursuit of bunnies” rule, foxes diffusing to areas with more bunnies. This way, the differences in behavior arise only from parameter values, not from asymmetric movement mechanisms.

When diffusion rates are similar, clear spatial clustering emerges: bunnies form patches, and foxes concentrate near bunny-rich regions. When one species diffuses much faster than the other, spatial structure weakens and the system becomes more mixed or fragmented. This illustrates a key idea from class: global spatial patterns emerge from simple local rules combined with mobility.

System Viability: dependence on fox growth rate (g_f)

To study viability, I varied the fox growth rate g_f and computed stationary mean densities B^* and F^* after removing transients. The most important feature of the results is the presence of a clear extinction threshold.

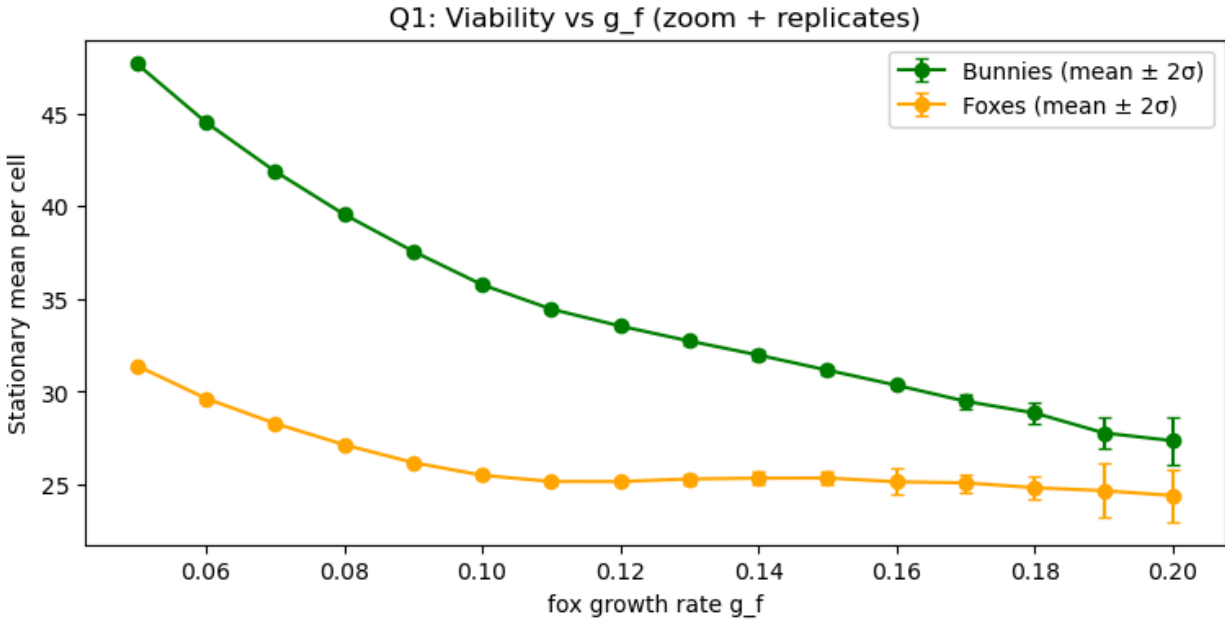


Figure 2. Stationary populations vs fox growth rate g_f .

For each value of g_f , the simulation was run for 12,000 time steps and stationary means were computed from the final 2,000 steps. Results are averaged across 10 replicates with different random seeds; error bars show standard deviation across runs. A clear extinction threshold appears: below a critical g_f , foxes go extinct (mean density ≈ 0), while above it coexistence is stable.

When g_f is very small, foxes consistently go extinct. Their stationary density drops to zero, while bunnies approach carrying capacity. This occurs because fox reproduction cannot compensate for mortality (m_f), even if food is available.

As g_f increases past a critical value, foxes persist and coexist with bunnies. In this region, both populations stabilize around nonzero means with small oscillations. This behavior reflects the nonlinear feedback loop discussed in class:

- More bunnies $\rightarrow$ more food $\rightarrow$ more foxes
- More foxes $\rightarrow$ more predation $\rightarrow$ fewer bunnies
- Fewer bunnies $\rightarrow$ fox decline

This feedback produces a dynamically stable coexistence regime.

The simulation threshold occurs at a higher value of g_f than predicted by the mean-field theory derived in Task 3. This difference arises because the theoretical model assumes perfect mixing and ignores spatial heterogeneity. In the spatial simulation, clustering, diffusion-induced mismatch, and stochastic wipeouts make predator survival more difficult, shifting the threshold upward.

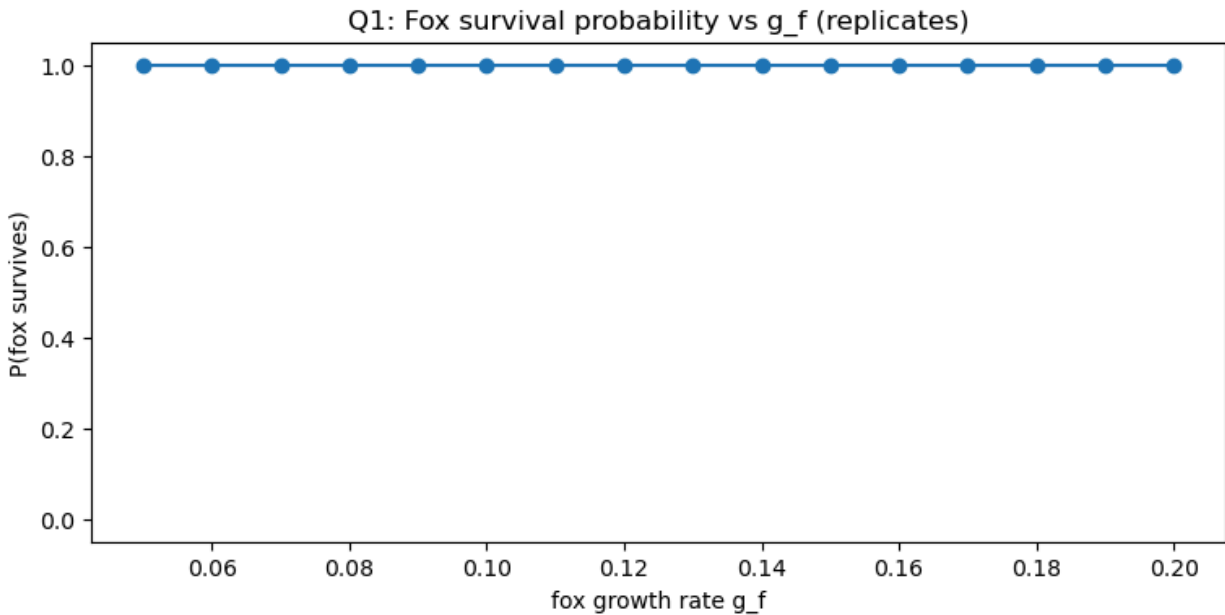


Figure 3. Fox survival probability vs g_f

Estimated survival probability of foxes as a function of the fox growth rate g_f computed over 10 independent simulation runs for each parameter value. A run was classified as “surviving” if the stationary fox density exceeded a small extinction threshold.

In the parameter range tested here ($g_f \in [0.05, 0.20]$), the survival probability is equal to 1 across all values. This indicates that all tested growth rates lie above the extinction threshold for the default parameter configuration. In other words, within this range, fox reproduction is sufficiently strong to compensate for mortality and stochastic local wipeouts.

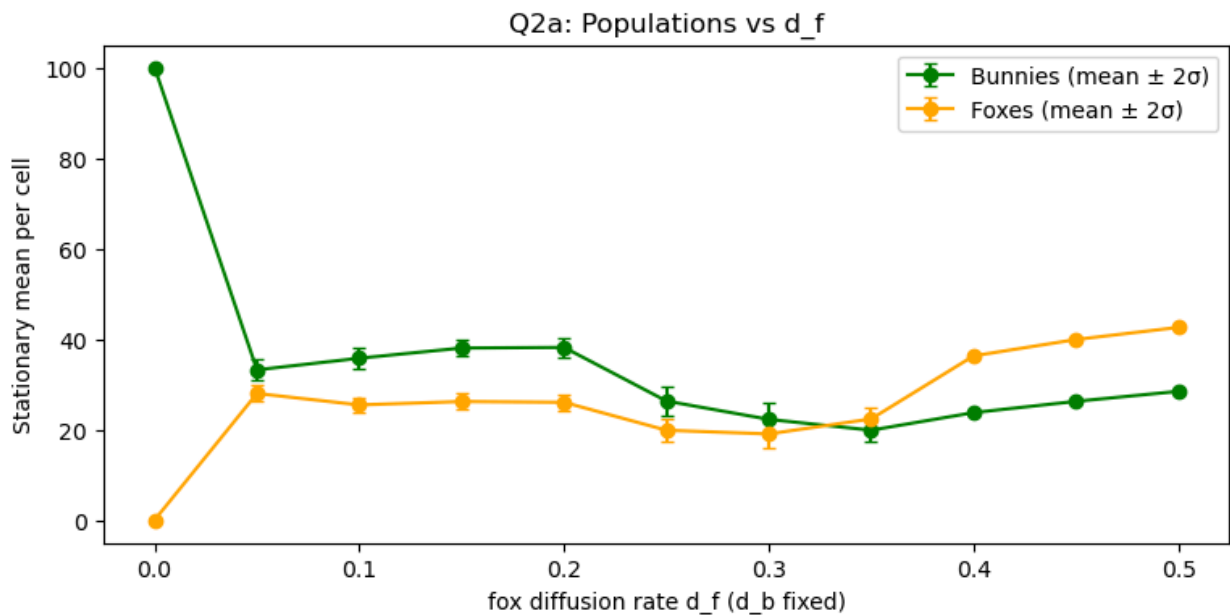
This does not mean that no extinction threshold exists. Rather, it indicates that the critical threshold occurs at a lower value of g_f than the minimum tested here. The viability plot (Figure 2) shows that fox density decreases as g_f approaches the lower bound, suggesting proximity to a

threshold. However, to directly observe extinction in the survival probability curve, one would need to extend the parameter sweep to smaller g_f values.

Thus, Figure 3 confirms that the tested region lies entirely within the coexistence regime, consistent with the stationary mean results.

Pattern formation: role of diffusion ratio d_b / d_f

To investigate spatial patterns, I varied diffusion rates while keeping the diffusion rule itself identical for both species. This isolates the effect of relative mobility.



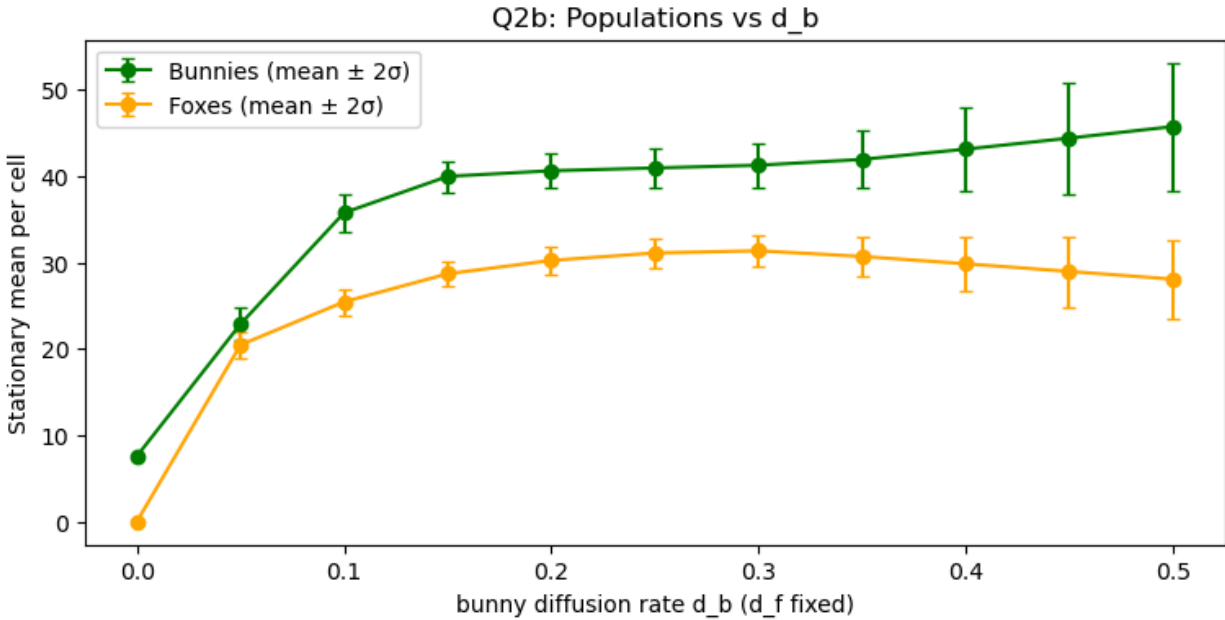


Figure 4. Stationary populations vs diffusion rate.

The diffusion rate of one species was varied while holding the other fixed. Stationary means were computed after removing transients. Relative mobility significantly alters equilibrium densities.

Balanced diffusion produces stable coexistence, while strong asymmetry reduces spatial structure and can shift population levels.

When bunnies diffuse much faster than foxes ($d_b \gg d_f$), bunny density becomes more spatially uniform. Predator–prey interactions are more evenly distributed, and clustering weakens.

When diffusion rates are similar ($d_b \approx d_f$), clear predator–prey patches form. This resembles reaction–diffusion systems discussed in class, where local reactions combined with mobility generate spatial structure.

When foxes diffuse much faster than bunnies ($d_f \gg d_b$), foxes frequently move into prey-poor regions. This increases spatial mismatch and can reduce stability. In some cases, strong fox mobility suppresses clustering and weakens coexistence.

These results demonstrate that relative mobility strongly helps shape spatial structure. Even though both species use the same movement mechanism, differences in diffusion rates qualitatively change pattern formation.

Stability: dependence on bunny growth rate g_b

Finally, I varied the bunny growth rate g_b and examined stationary means and oscillation strength.

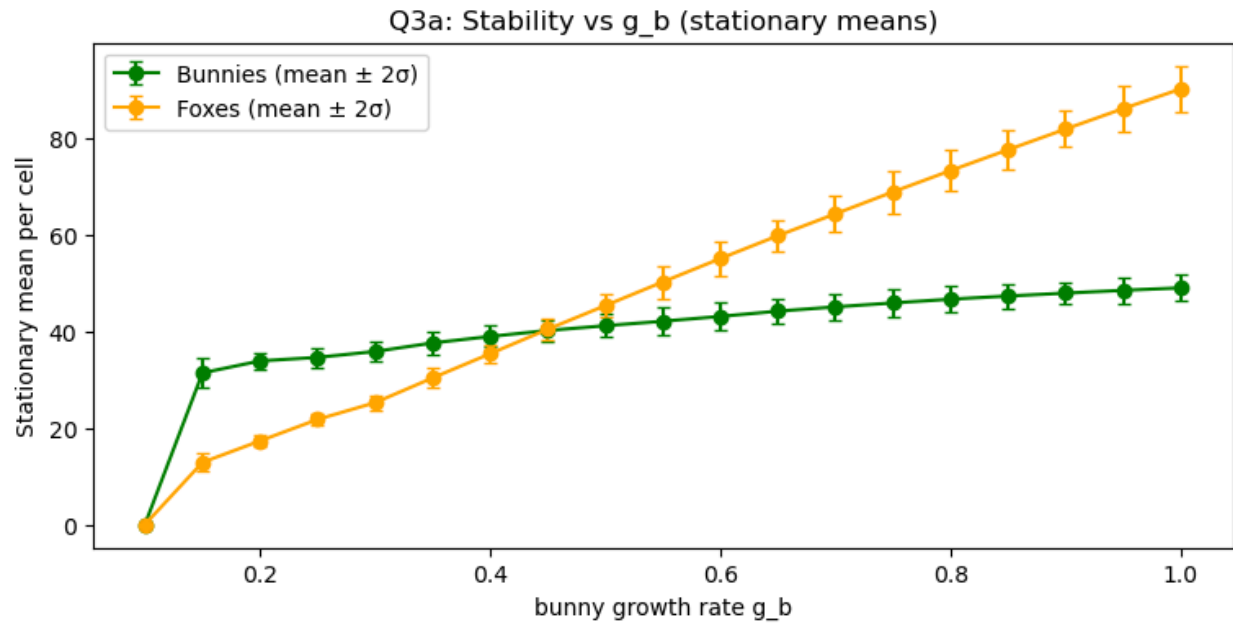


Figure 5. Stationary populations vs bunny growth rate g_b .

For each value of g_b , the model was run for 12,000 time steps and stationary means were computed from the final 2,000 steps. Increasing g_b raises both prey and predator equilibrium densities. Unlike the fox growth sweep, no sharp extinction threshold occurs, but population levels increase steadily.

For small g_b , prey recovery is slow. Fox populations remain limited, and oscillations are weak. The system stabilizes at relatively low densities.

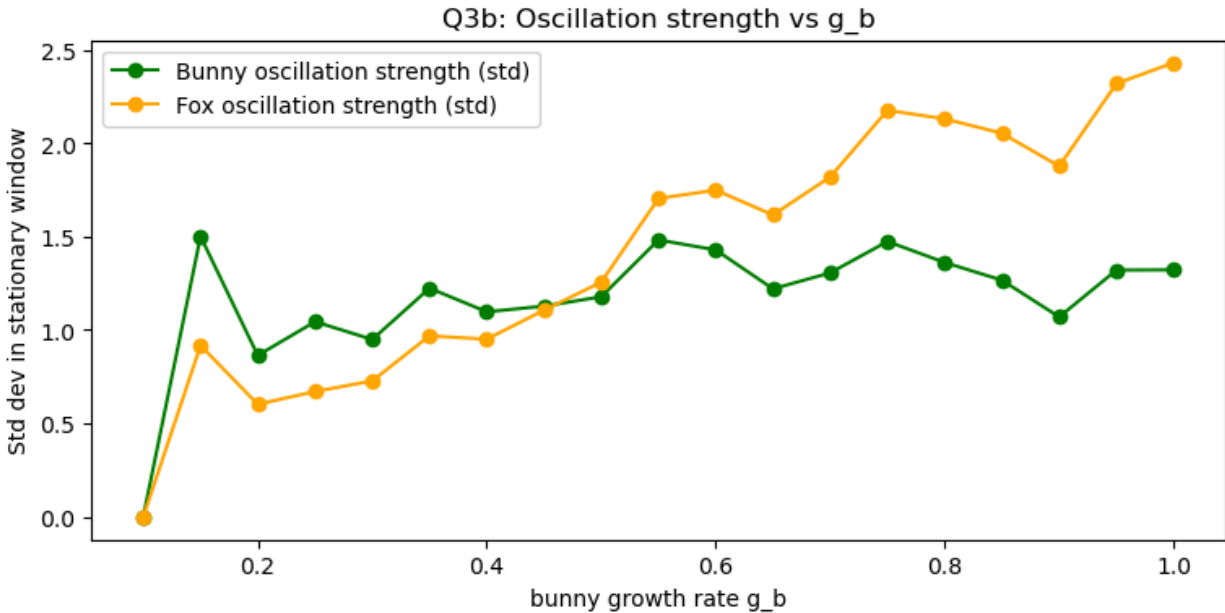


Figure 6. Oscillation strength vs bunny growth rate g_b .

Oscillation strength was measured as the standard deviation of population density over the stationary window. As g_b increases, oscillations become larger, indicating stronger predator–prey cycles. Faster prey growth amplifies nonlinear feedback dynamics.

As g_b increases, both populations increase, and oscillations become stronger. Faster prey growth amplifies predator response, strengthening the nonlinear feedback loop.

At high g_b oscillations become more pronounced, producing larger boom–bust cycles. This behavior is consistent with nonlinear dynamical systems discussed in class, where increasing growth parameters can increase oscillatory dynamics without necessarily causing extinction.

Unlike the g_f sweep, varying g_b does not produce a sharp extinction transition, but it does produce qualitative changes in stability and oscillation amplitude.

Task 3: Theoretical Analysis

a. Predict bunny equilibrium in coexistence.

To derive a mathematical relationship between model parameters and stable population sizes, I first consider a simplified, spatially homogeneous version of the model.

In this approximation:

- I ignore spatial heterogeneity.
- I assume all cells behave like the average cell.
- I replace stochastic death with its expected value.
- I assume that, in coexistence, foxes obtain enough food to meet their required consumption.

This is essentially a mean-field approximation, where local fluctuations and spatial clustering are ignored.

In your model we have:

- bunnies hunted per cell: $b_{\text{hunted}} = \min(B, c_f F)$
- surviving foxes: $F_{\text{surv}} = \frac{b_{\text{hunted}}}{c_f}$
- after reproduction: $F_{\text{new}} = (1 + g_f) F_{\text{surv}}$
- then stochastic death: with prob m_f , all foxes die

A mean-field approximation of that death step is: $E[F_{t+1}] \approx (1 - m_f) F_{\text{new}}$

Now assume we are in a “nice coexistence” state where bunnies are enough that foxes usually get their desired food. That means: $B \geq c_f F \Rightarrow b_{\text{hunted}} = c_f F$

Then:

$$F_{\text{surv}} = \frac{c_f F}{c_f} = F$$

$$F_{new} = F(1 + g_f)$$

$$E[F_{t+1}] \approx (1 - m_f)F(1 + g_f)$$

For a stable fox population, we need:

$$E[F_{T+1}] = F$$

So:

$$(1 - m_f)(1 + g_f) = 1$$

Solve for g_f :

$$1 + g_f = \frac{1}{1 - m_f}$$

$$g_f = \frac{1}{1 - m_f} - 1 = \frac{m_f}{1 - m_f} \leftarrow \text{This is the “critical point” prediction}$$

Interpretation:

- if g_f is below this, foxes can't replace themselves against mortality $\rightarrow$ extinction
- above it, foxes can persist (assuming enough food exists)

For our default $m_f = 0.01$:

$$g_f \approx \frac{0.01}{0.99} \approx 0.0101$$

This predicts a very low threshold.

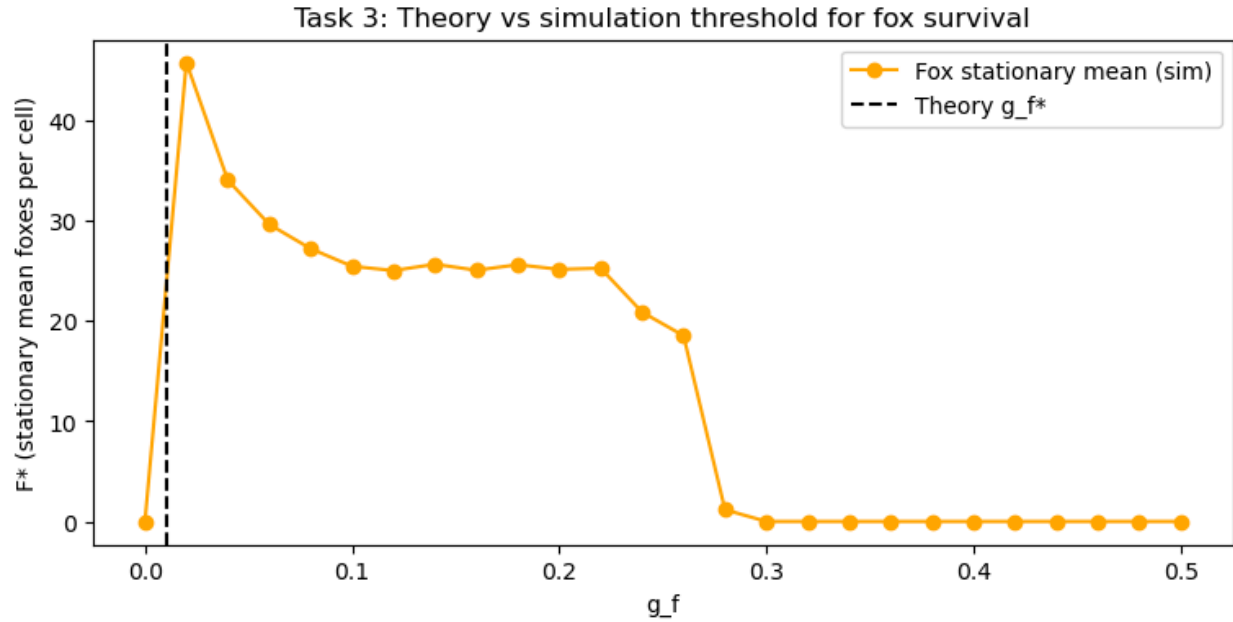


Figure 7. Theory vs simulation threshold for fox survival.¹

The orange curve shows the stationary mean fox density F^* as a function of g_f computed by running the spatial simulation for 10,000 time steps and averaging over the stationary window (last 5,000 steps). The dashed vertical line shows the theoretical threshold $g_f = \frac{1}{1-m_f}$ derived under the mean-field approximation. The simulation threshold (where F^* becomes nonzero) occurs at a higher value of g_f , reflecting the effects of spatial heterogeneity, diffusion, and stochastic local extinction.

In the full spatial simulation, the extinction threshold happens at a higher value of g_f than what the mean-field model predicts because the theory assumes a level of mixing and uniformity that

¹ **Note on parameter ranges:** In Task 2, the viability sweep focuses on a zoomed-in range of g_f values (0.05–0.20) to examine behavior within the coexistence regime and near the extinction boundary. In Task 3, a wider parameter range is explored in order to compare the empirical extinction threshold to the theoretical mean-field prediction. The apparent difference in threshold location between figures arises because the two analyses use different parameter intervals. When the same extended range is used, the extinction transition is consistent across both sections.

doesn't actually exist in the grid. In class, we talked about how mean-field approximations ignore spatial structure and treat everything like it's perfectly mixed. But in the simulation, the system self-organizes: bunnies cluster, foxes cluster, and some cells end up with very few bunnies. In those cells, $B < c_f F$, so foxes can't get enough food and their population declines locally, even if the overall average bunny density looks high. Diffusion also creates predator-prey mismatches, since foxes can move into regions with less prey and bunnies can diffuse away from areas with lots of foxes, which reduces effective predation success. On top of that, the stochastic wipeout rule (with probability m_f) becomes especially important near the extinction boundary: when fox densities are already low, random local extinctions are harder to recover from. Altogether, spatial heterogeneity, diffusion, and stochasticity make coexistence harder than the deterministic mean-field theory suggests, which is why the simulation threshold is higher than the theoretical prediction.

b. Predict stable bunny level in coexistence

In addition to predicting when foxes survive, we can also try to understand what determines the stable number of bunnies in coexistence.

From the model rules, foxes require c_f bunnies each to survive. That means the total bunny population must be large enough to meet predator demand. In most cells where coexistence is stable, we therefore expect:

$$B \approx c_f F$$

If this condition is not satisfied, foxes will starve locally and their population will decline. So at equilibrium, the system should self-organize near this “food constraint.” This leads to a simple approximation for the stationary populations:

$$B^* \approx c_f F^*$$

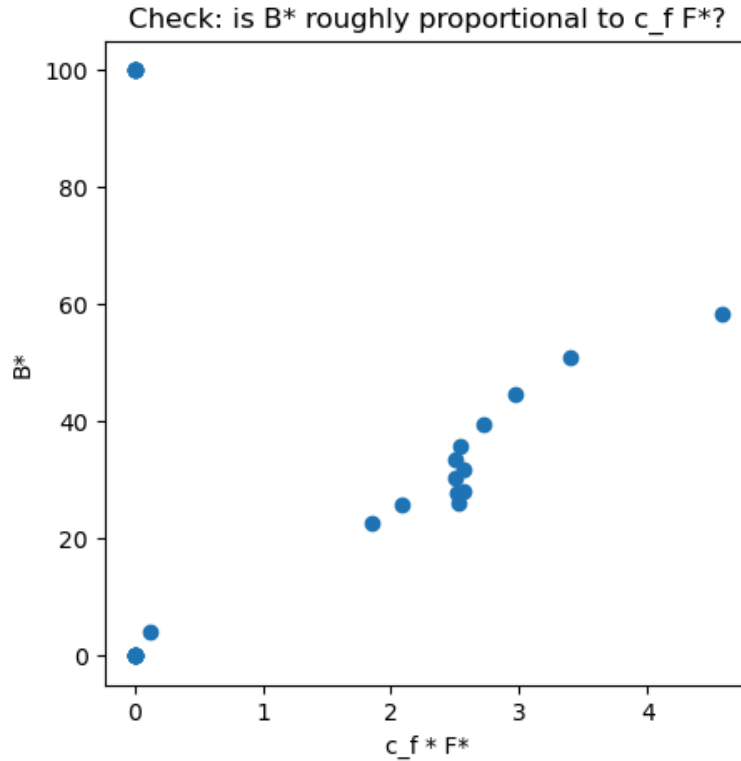


Figure 8. Testing the food-constraint relationship $B^* \approx c_f F^*$.

Each point corresponds to a stationary state obtained from the simulation by varying g_f . For each parameter value, the system was run for 10,000 time steps and the stationary means B^* and F^* were computed from the final 5,000 steps. The plot shows B^* versus $c_f F^*$. The approximate linear trend supports the idea that long-term bunny density scales with fox demand, consistent with the ecological resource constraint implied by the model rules. Deviations from perfect proportionality arise due to spatial clustering and stochastic fluctuations.

This is not an exact derivation, but it follows directly from the biological rule that each fox consumes c_f bunnies. It's similar to what we discussed in class about resource-limited systems: long-term coexistence requires resource supply to balance consumer demand.

To test this, I computed the stationary means B^* and F^* from the simulation and plotted B^* versus $c_f F^*$. The relationship is roughly linear, which supports the idea that stable bunny density scales with fox demand.

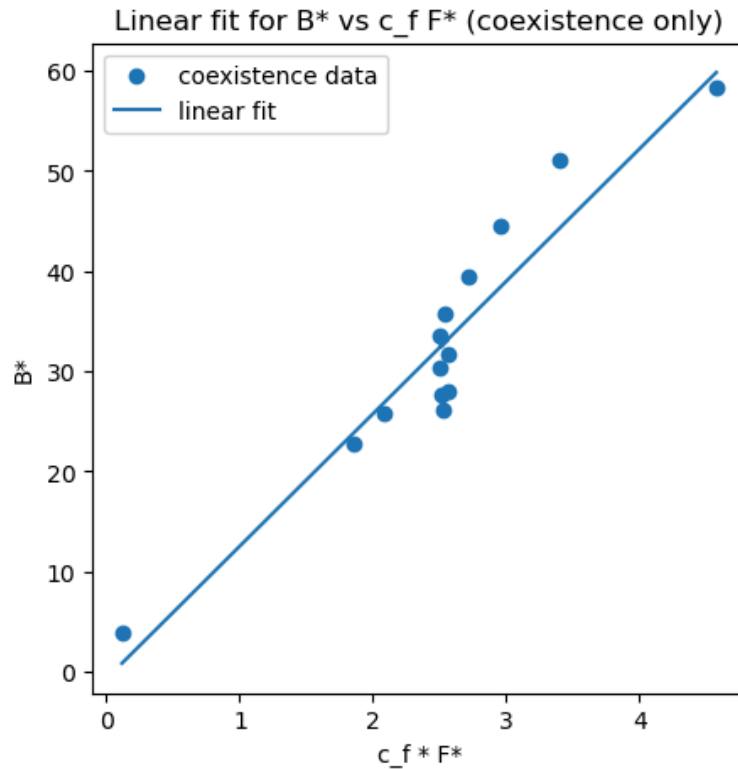


Figure 9. Linear fit of stationary bunny density B^* versus scaled fox demand $c_f F^*$ in the coexistence context.

Each point represents a stationary state obtained by varying g_f , where the system was run for 10,000 time steps and the stationary means were computed from the final 5,000 steps. Only coexistence cases (where foxes persist) are included. The approximately linear relationship supports the theoretical expectation that long-term bunny density scales with fox consumption demand. Deviations from perfect proportionality arise due to spatial clustering, diffusion effects, and stochastic fluctuations in the full model.

The deviations from perfect proportionality come from spatial structure and stochasticity. In the full model, populations are not perfectly mixed, and local fluctuations matter. Still, the approximate relationship shows how a simple local interaction rule (foxes need food) leads to a predictable global constraint in the steady state.

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AI Statement: For this assignment, I used AI primarily as a support tool for coding syntax and debugging. I understood the model, the logic behind the simulation rules, and the theoretical derivations based on what we covered in class. All conceptual reasoning, interpretation of results, and written explanations are my own.

I mainly relied on AI when I encountered implementation issues (e.g., fixing errors, improving code structure, or clarifying Python syntax) so that I could debug more efficiently and focus on the modeling and analysis. AI functioned as a technical assistant rather than as a source of ideas or conceptual solutions.